

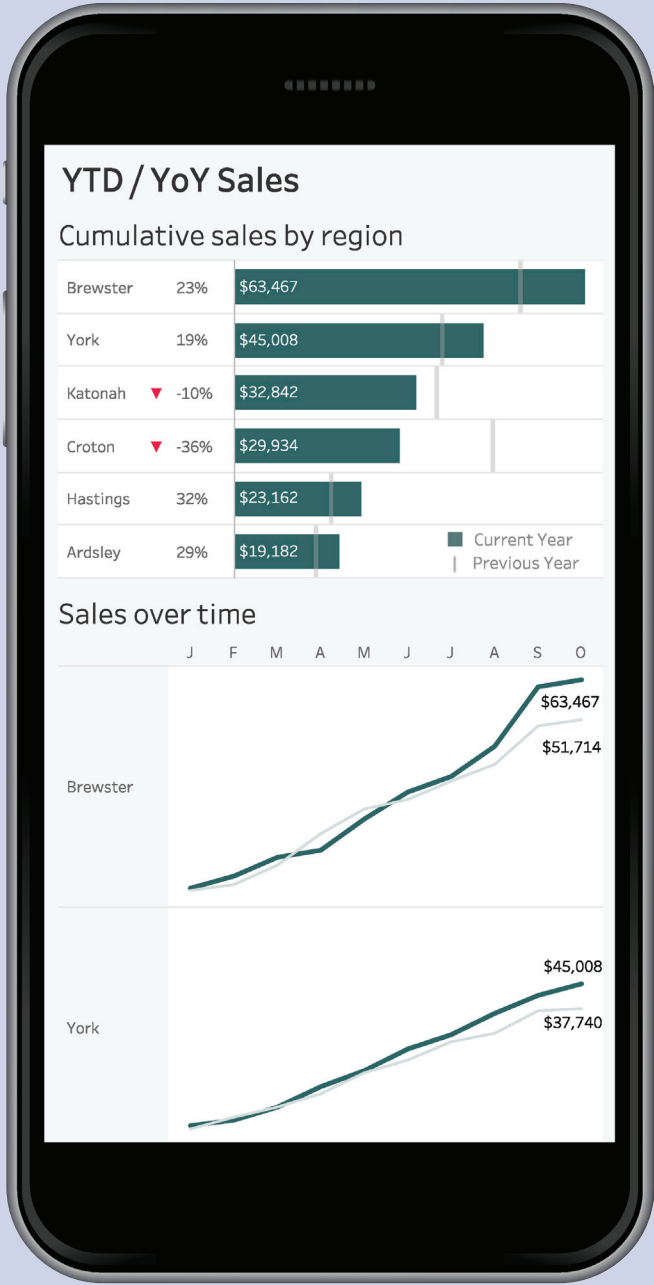
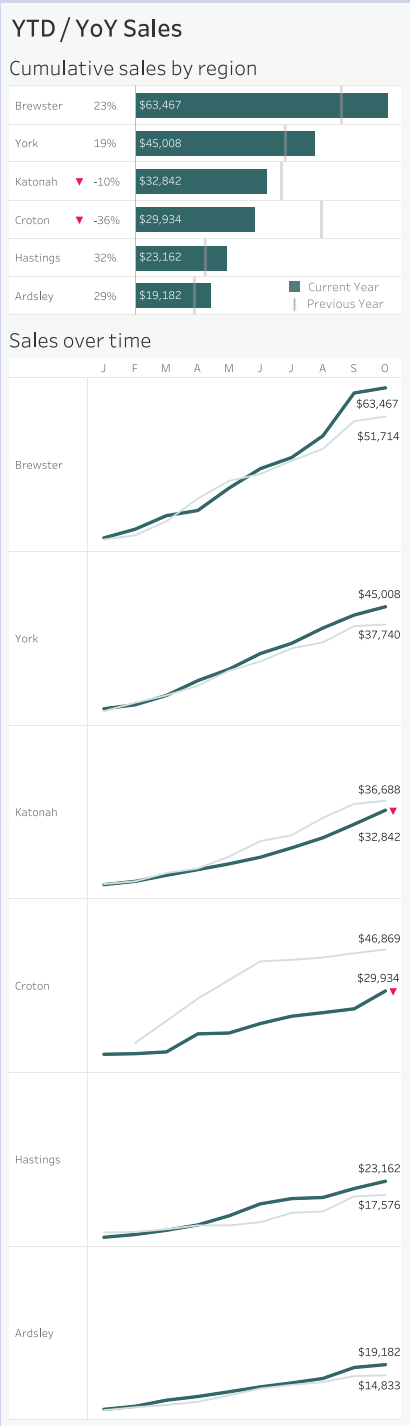
Chapter 10

Showing Year-to-Date and Year-over-Year at the Same Time

Year-to-date/Year-over-year dashboard optimized for mobile devices.

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Organization: Data Revelations



Note

Depending on your needs, you may want to go with the alternative approach suggested by Andy Cotgreave at the end of this chapter.

Scenario

Big Picture

You received a promotion in June of this year that involves managing sales for six different regions. Your number-one goal is to make sure that by the end of the year, the sales for each region will be greater than they were for the previous year.

You came into the new job knowing that two of the regions were way behind where they needed to be. In your discussions with upper management, you made the argument that a reasonable goal would be to exceed sales in four regions and improve the trend in the other two regions that were tanking.

As you are often out of the office visiting individual stores, you want to have a collection of mobile-friendly dashboards that allow you to monitor current performance, trends, and near-real-time details on demand. Back at the office, you have a single dashboard that combines all these elements into one view.

Here we'll look at how to fashion a dashboard that makes it easy to compare performance across regions and years and also detect trends.

Specifics

- You need to rank year-to-date (YTD) sales performance for six regions and compare with

sales for the previous year (i.e., year-over-year [YoY]).

- You need to see, at a glance, how much better or worse the regions are performing.
- You need to see trends to know whether you are on track to hit your goals.
- You need a way to view sales activities for a particular region and for a particular time period.

Related Scenarios

- You need to monitor first-call resolution rates for several different products and compare them with rates from the previous year.
- You are the head of annual giving for a university and need to rank and compare donations for 10 different regions for this year and for the previous year.
- You monitor attendance for theatrical productions and need to compare current YTD ticket sales with the previous year's sales.
- In this book, check out these chapters: Chapter 5, which uses an index chart to compare quarter-to-date performance; Chapter 6, which compares one quarter to another; Chapter 7, which shows if you are ahead of or behind where you need to be to reach your goals; and Chapter 8, which compares current performance against goals.

How People Use the Dashboard

Opening the dashboard shows a summary of the six regions as of today. (See Figure 10.1.)

The key performance indicator (KPI) icons show there are problems in Katonah and Croton, so the user can scroll down and look at trends for those two regions. (See Figure 10.2.)

YTD / YoY Sales

Cumulative sales by region

Brewster	23%	\$63,467	
York	19%	\$45,008	
Katonah	▼ -10%	\$32,842	
Croton	▼ -36%	\$29,934	
Hastings	32%	\$23,162	
Ardsley	29%	\$19,182	
		Current Year	Previous Year

Sales over time

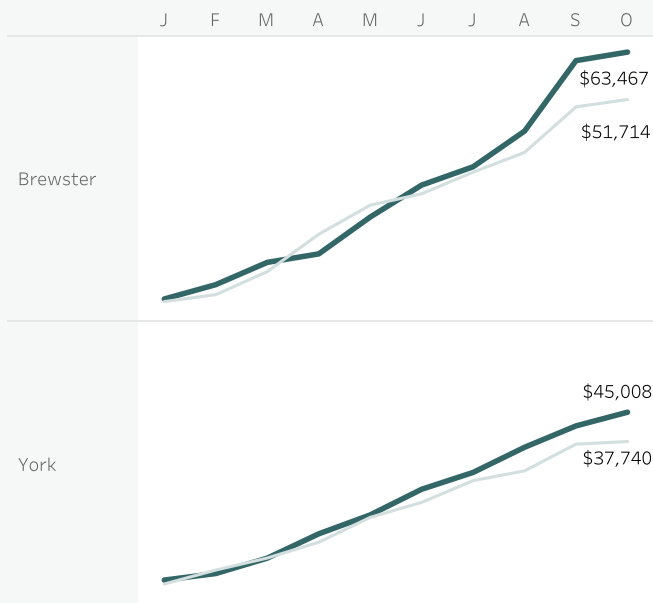


FIGURE 10.1 Initial dashboard view showing summary and trends for first two regions.

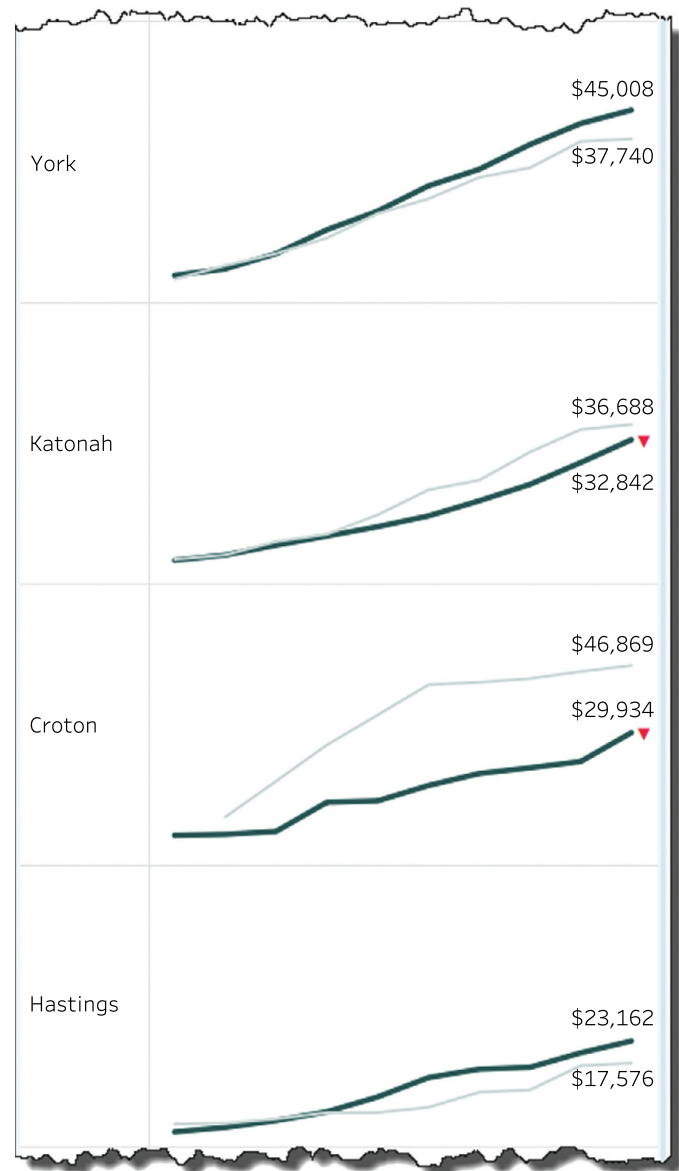


FIGURE 10.2 Scrolling allows the user to look at the regions that are underperforming.

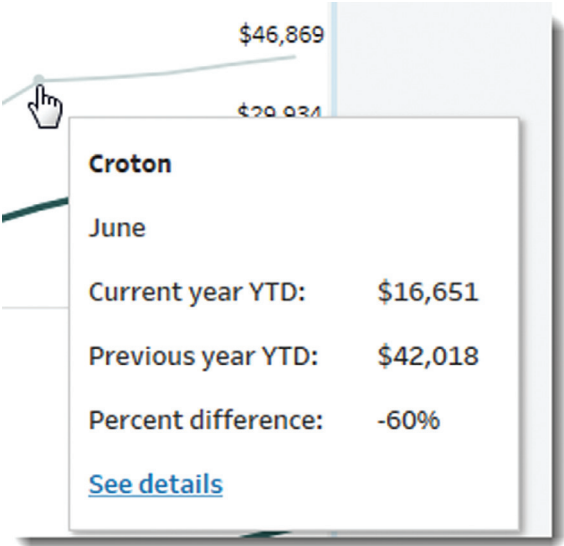


FIGURE 10.3 Pressing a point shows details about activity for that month.

The good news here is that while YTD sales are down as of today, things are much better now than they were a few months ago when you took the job. You can place your finger over a data point and see information about that time period. (See Figure 10.3.)

If you want to see sales details for the region and period in question, you can press the “See details” link. Examples of what you might see would be sales by day of the week and for individual items.

Why This Works

Bars Allow for Easy Comparison

The bar chart makes it very easy to see just how much greater one region’s sales are compared with another’s. (See Figure 10.4.)

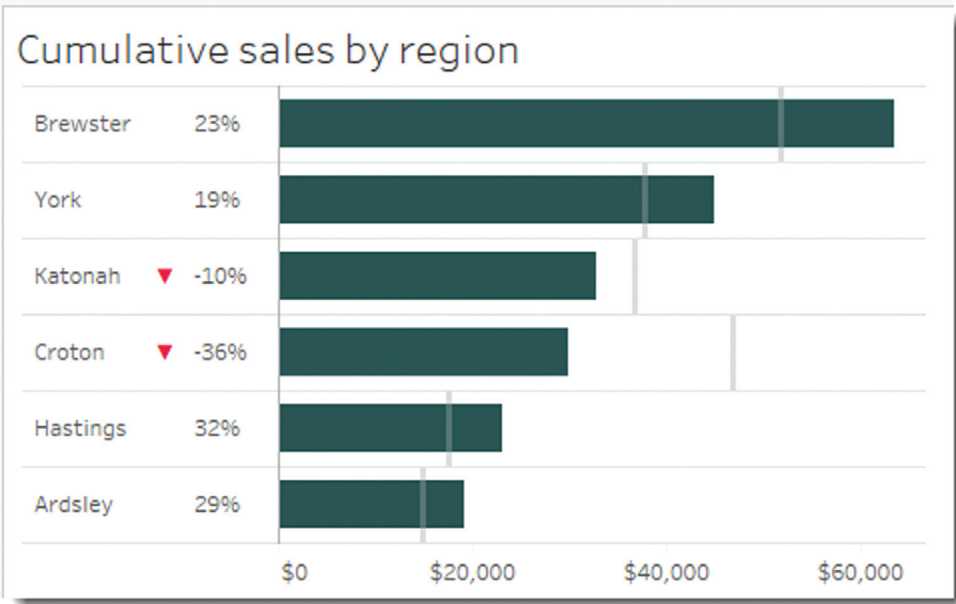


FIGURE 10.4 Bars make it easy to see how much larger one region’s sales are compared with another’s.

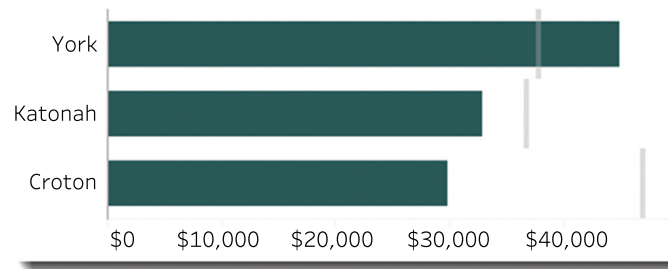


FIGURE 10.5 Reference lines make it easy to see how far ahead or behind we are.

Indeed, we could leave the labels that are inside the bars off the visualization, and it would still be easy to tell that sales for Brewster are about twice those of Katonah.

Reference Lines Make It Easy to See How Far Ahead or Behind We Are

Again, we can remove the numbers and the KPI icons but still see that Croton is significantly behind the

previous year while Katonah is only a little behind and York is quite a bit ahead. (See Figure 10.5.)

Good Use of Color and Good Placement of Color Legend

We show the current year in a dark green and the previous year in light gray, as the current year is our primary concern. (See Figure 10.6.) The color and shape legend (which users don't even need at this point because they've looked at this dashboard hundreds of times) is tucked inside the chart and does not take up any additional screen space.

Although the KPI icon is red, the very existence of the icon and the fact that it is pointed down alerts all users (even those with color vision deficiency) that there is a problem in Katonah and Croton.

Number Placement Optimizes Space

Given that the target device is a smartphone, there's not a lot of room for the bars. Placing the numbers

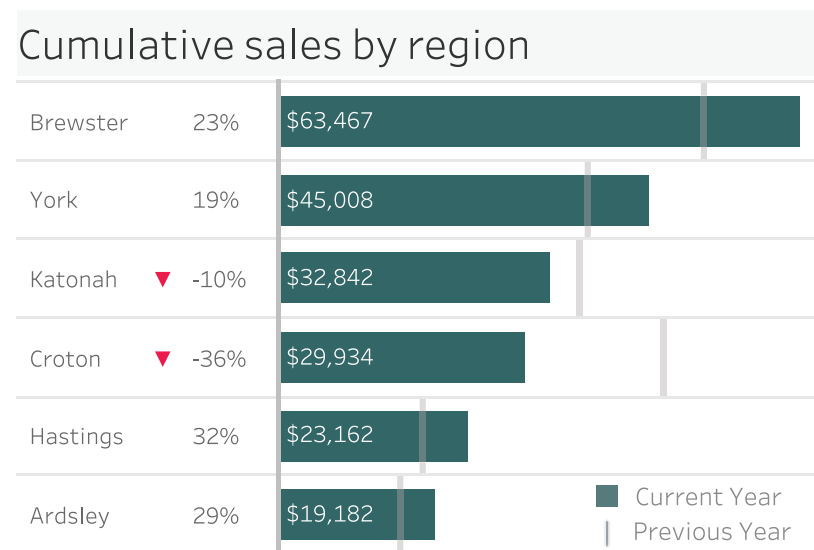


FIGURE 10.6 Colors, icons, and legend placement make it easy to decode the chart.

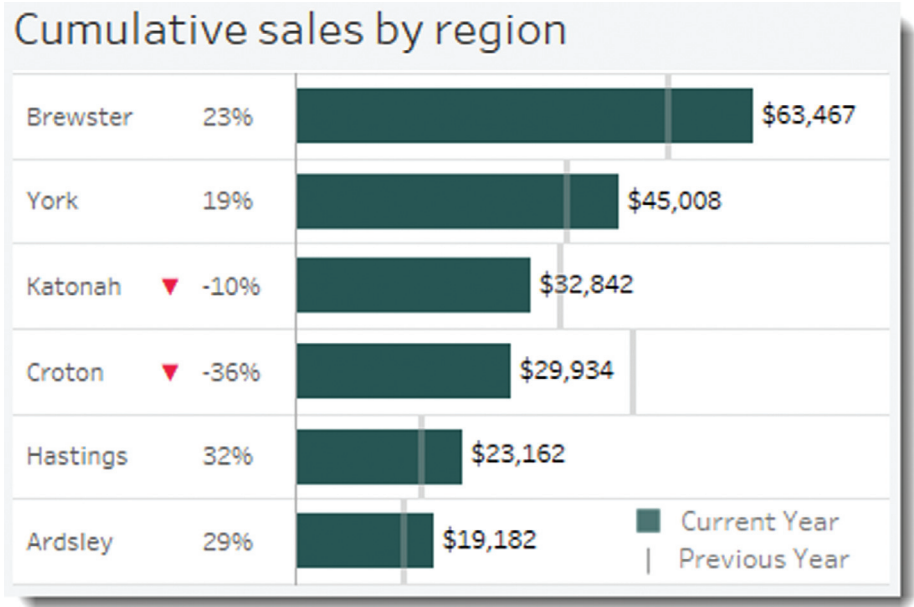


FIGURE 10.7 Numbers placed outside the bars cramp the chart and conflict with the reference lines.

inside the bars gives the chart more space and avoids a conflict with the reference lines.

Figure 10.7 shows what the chart would look like if the numbers were placed outside the bars.

Line Charts Make It Easy to Glean Trends

Although it’s great to have a quick summary view, it’s important to know how things are trending. For example, Katonah is down 10 percent from the previous year. Did that gap happen recently, or has there been a problem from the beginning of the year? The trend line in Figure 10.8 makes it easy to see

that the gap was much larger earlier in the year and that the sales for the region have made up a lot of ground from the previous year.

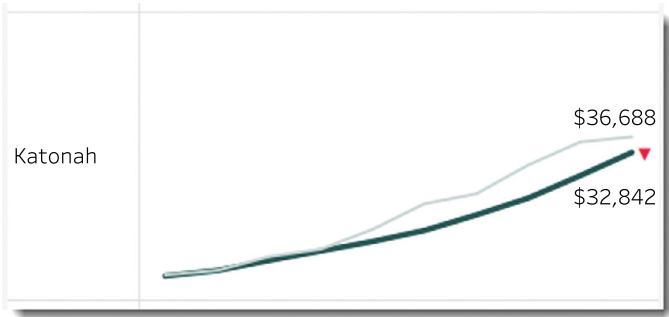


FIGURE 10.8 Trend lines show that the gap is narrowing.

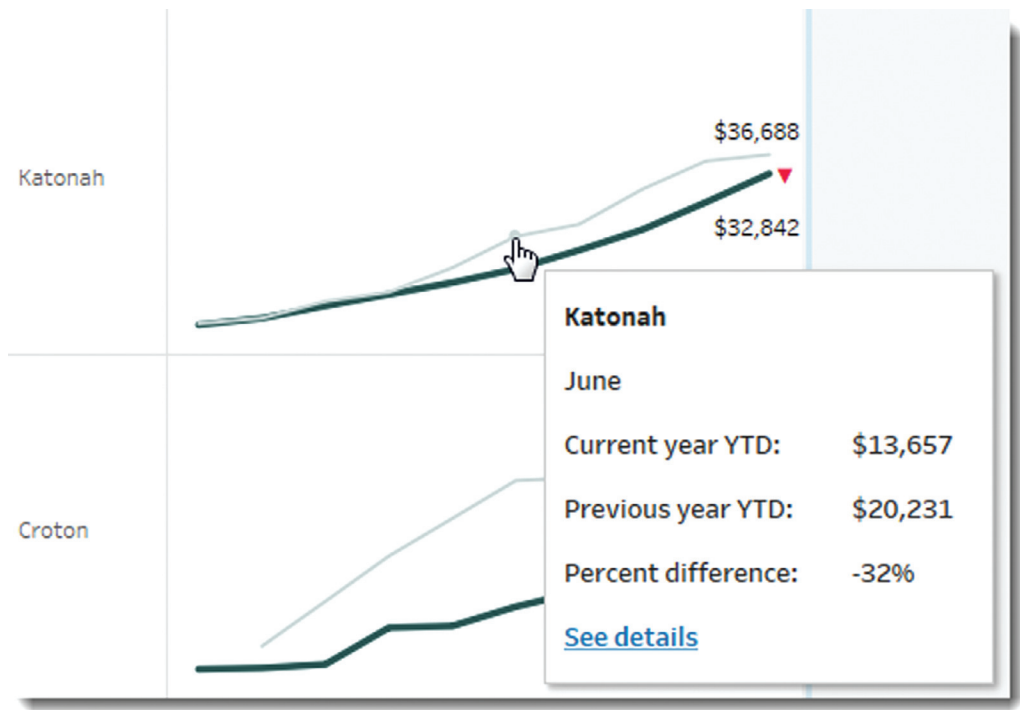


FIGURE 10.9 Back in June, current YTD sales trailed the previous year by 32 percent.



Selection Allows Easy Exploration of Different Time Periods and Details on Demand

The user can press any point and get additional information about that point. In Figure 10.9, we see that the sales gap in June was quite a bit larger than it is now.

The “See details” hyperlink allows users to see sales details about the selected period. The link could jump to the actual sales system or to a dashboard with details about the month and region in question.

The Traditional Approach, and Why You Should Avoid It

Figure 10.10 exemplifies the tabular scorecard approach to displaying YTD and YoY data.

Cumulative sales by region

		Current Sales	Previous Sales	% Difference
Brewster		\$63,467	\$51,714	23%
York		\$45,008	\$37,740	19%
Katonah	▼	\$32,842	\$36,688	-10%
Croton	▼	\$29,934	\$46,869	-36%
Hastings		\$23,162	\$17,576	32%
Ardsley		\$19,182	\$14,833	29%

FIGURE 10.10 A typical scorecard approach to displaying YTD and YoY data.

The table with the KPI icons is not aesthetically unpleasing; it just makes us work very hard to understand the magnitude and differences in the numbers. For example, Brewster has sales of \$63,467 and Ardsley is at \$19,182. What are users to make of this? Users have to do mental math to understand that the region with the most sales is more than three times as much as the region with the least sales.

The same goes with the percent difference column. Users must do mental math and imagine bars of different lengths. Compare this with Figure 10.5, where users don't have to work nearly as hard to really understand the difference in magnitude.

Dashboard Designer Commentary

Steve: The reason I wanted to include this example was that I have seen so many companies struggle to show YoY performance. The chart in Figure 10.10 typifies their efforts.

I admit that I may have left some important considerations out of the scenario. For example, if this were a desktop dashboard, I would almost certainly show the sales details on it.

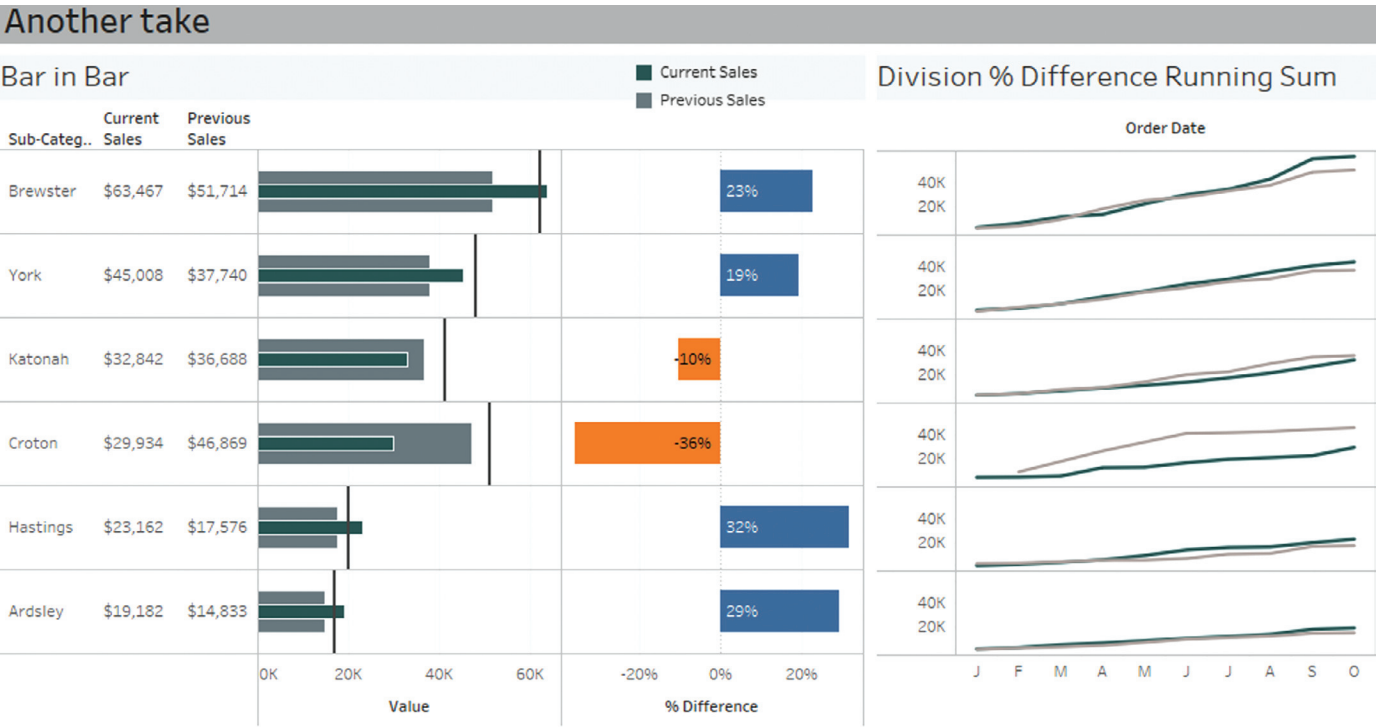


FIGURE 10.11 An early attempt to compare current-year YTD sales and prior-year YTD sales along with progress toward a goal.

And suppose, in addition to showing performance this year versus performance last year, I wanted to also show a goal for each region. Sure, we want to have better sales this year than last year, but we don't just want it to be better; we want it to be at least 20 percent better in each region. How can we show that too?

Figure 10.11 is a very early iteration of such a dashboard where I present a bar-in-bar chart showing current-year versus prior-year sales and a reference line showing a goal.

Although it is a good first effort, this chart is nowhere near ready to go into production. Between the bar-in-bar chart and the reference line, there's a lot for the user to parse. And I can't tell if the blue and orange bars showing percentage difference compare current progress with last year or with the goal.

I also placed the trend lines to the right of the bar chart in an attempt to make it easy to see trends for each region. The problem here is that the height for each row restricts the ability to see all but the most obvious gaps.

If I had a client who needed all this functionality, I would revisit this approach, but for the particular scenario I established, I think the simpler, mobile-friendly dashboard does the job.

Author Commentary

Which Chart to Choose?

ANDY: When tracking YoY performance, you often have to consider two options:

Do you visualize both years' actual sales on top of each other?



FIGURE 10.12 A version of the dashboard showing the YoY difference rather than cumulative sales. Note that each axis in the area chart is independent.

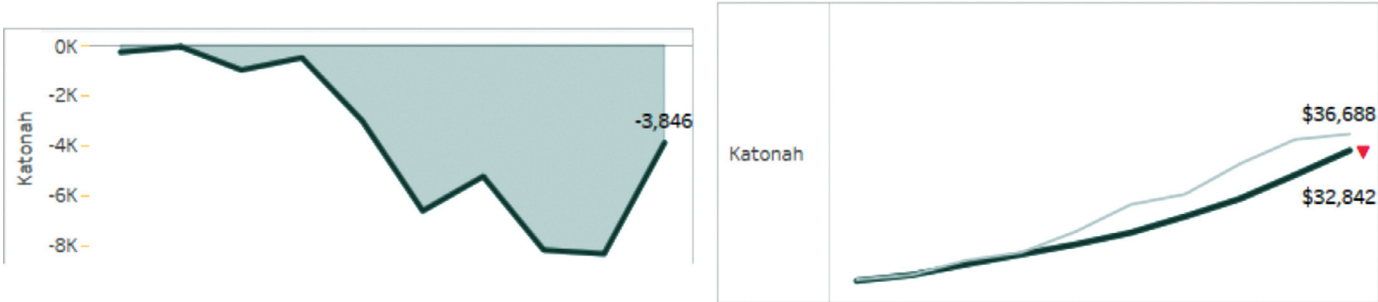


FIGURE 10.13 Two approaches to showing YoY sales for Katonah.

Do you visualize the actual difference itself?

Both methods have pros and cons. In this case, Steve chose the former. In his example, you see the actual number represented on the line. You can see whether the gap is large or small between the years, but it's not easy to visually measure the gap precisely.

That got me thinking: What does this look like if we visualize the actual difference itself? (See Figure 10.12.)

Now we see the difference with absolute clarity. Remember that problem we have in Katonah and Croton? You can see just how bad that is. The YoY differences have gotten worse in almost all months this year. The most recent month has seen an improvement, but there is still some way to go to achieve the goal.

In Figure 10.13, you can compare the different approaches. If knowing the exact YoY difference is most important, I recommend using the approach on the left. The downside of this approach is that

you can no longer see the *actual* cumulative sales. That's a reason to go with the approach on the right.

Could you also build a dashboard that visualizes both actuals and the difference? You could, but then you begin to complicate the dashboard, which could mean the insight is lost and, ultimately, people don't use it, because it's not simple enough. One compromise would be to have a switch that allows the person interacting to toggle back and forth between the two views. (See Figure 10.14.)

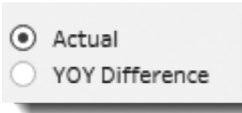


FIGURE 10.14 A simple switch at the top of the dashboard allows users to toggle back and forth between the two views.

Bars or Tables

Steve rightly compares the efficiency of bar charts and tables in Figures 10.5 and 10.10. Bar charts with reference lines are without doubt a more efficient

way to accurately compare values in different categories. However, I will put forward a defense of tables, lest you think we're advocating you never put a table on your dashboard.

If the purpose is to compare many values at a glance then a chart trumps a table all the time.

However, *if* you need to look up exact values *and* fast comparison is not the primary task, then a table or highlight table might be best. Our visual system is amazing, which is why we visualize data, but do be mindful that there are circumstances where it is more important to know the exact number than anything else.